

Euclid's Elements — Book V

Proposition 14

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If a first magnitude has to a second the same ratio as a third to a fourth, then the comparison of the first with the third is the same as the comparison of the second with the fourth.

1 Introduction

Proposition V.14 is the central comparison theorem of the present Book V reconstruction.

Given equal ratios

$$a : b = c : d,$$

it states that the comparison between the first terms a, c agrees with the comparison between the second terms b, d . Thus strict inequality, equality, and reverse strict inequality are all transported across the proportion.

In the formal development V.14 is not treated as an isolated ratio identity. Its strict direction is built from the already reconstructed propositions V.8, V.13, and V.10. The equality direction is then obtained by trichotomy and incompatibility between ratio equality and strict greater-ratio comparison.

The production file is

```
CGJteamLab/Proposition5_14.lean
```

and it imports

```
CGJteamLab.Proposition5_8  
CGJteamLab.Proposition5_10  
CGJteamLab.Proposition5_13
```

so that the proposition-level dependency graph is explicit at the module level.

2 Formal Statement of Proposition V.14

Assume

$$a : b = c : d.$$

The production theorem packages three equivalences:

$$a < c \iff b < d,$$

$$a = c \iff b = d,$$

and

$$c < a \iff d < b.$$

These are the three possible comparisons supplied by trichotomy on positive segment classes.

The production theorem is therefore stronger than a single implication: it records complete comparison preservation between corresponding terms of equal ratios.

3 Why V.14 Is Structurally Important

V.14 is the theorem that turns ratio equality into order information about individual magnitudes.

This is exactly the kind of step needed later in V.16. There, after scaling two equal ratios by arbitrary positive multiples, V.14 transfers the comparison between the scaled first terms to the scaled second terms.

In the classical route to XI.23, V.16 will then allow an inequality such as

$$OP < OL$$

to be transported to the corresponding inequality

$$PQ < LM.$$

Thus V.14 is not merely an internal Book V convenience. It is one of the main order-transfer engines of the later geometry.

4 Strict Direction: $a < c \Rightarrow b < d$

The core proposition-local theorem is

```
hilbertEudoxus_v14_lt_second_of_lt_first
```

with hypotheses

$$a : b = c : d$$

and

$$a < c.$$

The proof has exactly three Euclid-numbered steps.

4.1 Step 1: V.8

From

$$a < c$$

V.8 gives, with the common second term d ,

$$c : d > a : d.$$

In Lean this is

```
hilbertEudoxus_v8_greater_first_same_second
```

applied to c, a, d .

4.2 Step 2: V.13

The given equality

$$a : b = c : d$$

transports the strict comparison

$$c : d > a : d$$

to

$$a : b > a : d.$$

This is exactly Proposition V.13.

4.3 Step 3: V.10

Now the two ratios

$$a : b \quad \text{and} \quad a : d$$

have the same first term. V.10 reflects their strict comparison back to the second magnitudes:

$$a : b > a : d \implies b < d.$$

Hence

$$a < c \implies b < d.$$

5 Strict-Direction Dependency DAG

The formal and classical proposition-level graph coincide:

$$\begin{array}{c} a < c \\ \Downarrow \text{V.8} \\ c : d > a : d \\ a : b = c : d \\ \Downarrow \text{V.13} \\ a : b > a : d \\ \Downarrow \text{V.10} \\ b < d. \end{array}$$

This is the most important dependency chain in the Book V reconstruction up to this point.

6 Equality Direction: $a = c \Rightarrow b = d$

The proposition-local theorem is

```
hilbertEudoxus_v14_eq_second_of_eq_first
```

Assume

$$a : b = c : d$$

and

$$a = c.$$

After substitution the proportion becomes

$$a : b = a : d.$$

Now apply trichotomy to b, d :

$$b < d, \quad b = d, \quad d < b.$$

The middle case is the desired conclusion.

6.1 Case $b < d$

V.8 gives

$$a : b > a : d.$$

But we also have

$$a : b = a : d.$$

The interface theorem

```
hilbertEudoxusProportion_not_greater
```

states that Eudoxus-equal ratios cannot simultaneously stand in the greater-ratio relation.

Hence $b < d$ is impossible.

6.2 Case $d < b$

V.8 gives

$$a : d > a : b.$$

The original equality of ratios is first reversed using

```
hilbertEudoxusProportion_symm
```

to obtain

$$a : d = a : b.$$

Again equality of ratios excludes a simultaneous strict greater-ratio comparison.

Thus $d < b$ is impossible.

Only

$$b = d$$

remains.

7 The Full Less-Than Equivalence

The theorem

```
hilbertEudoxus_v14_lt_iff
```

states

$$a < c \iff b < d.$$

The forward direction is exactly the strict theorem already proved.

For the reverse direction assume

$$b < d$$

and apply trichotomy to a, c .

If

$$a < c,$$

we are done.

If

$$a = c,$$

the equality direction gives

$$b = d,$$

contradicting $b < d$.

If

$$c < a,$$

symmetry of the original proportion gives

$$c : d = a : b.$$

Applying the already established strict direction to $c < a$ then yields

$$d < b,$$

contradicting $b < d$.

Thus only

$$a < c$$

is possible.

8 The Full Equality Equivalence

The theorem

```
hilbertEudoxus_v14_eq_iff
```

states

$$a = c \iff b = d.$$

The forward direction is the equality theorem already proved.

The reverse direction is more delicate. Assume

$$b = d$$

and apply trichotomy to a, c .

If

$$a < c,$$

then the less-than equivalence gives

$$b < d,$$

contradicting $b = d$.

If

$$a = c,$$

we are done.

If

$$c < a,$$

reverse the proportion and use the less-than equivalence to obtain

$$d < b,$$

again contradicting $b = d$.

Therefore

$$a = c.$$

9 A Formalization Correction in the Equality Proof

The reverse implication

$$b = d \implies a = c$$

exposed an important directionality issue during compilation.

An earlier attempted proof tried to reverse the proportion and then reuse the lemma

$$\text{equality of first terms} \implies \text{equality of second terms}.$$

But after reversing

$$a : b = c : d$$

to

$$c : d = a : b,$$

the hypothesis $b = d$ is still an equality of the second terms, not of the first terms. The lemma therefore cannot be applied in that direction.

The corrected proof uses trichotomy on a, c and eliminates both strict cases via the already proved less-than equivalence.

This correction is mathematically informative: V.14 equality is not obtained by a naive symmetry of the one-way equality lemma. The comparison theorem is needed.

10 Reverse Strict Comparison

The theorem

```
hilbertEudoxus_v14_gt_iff
```

states

$$c < a \iff d < b.$$

No new argument is required.

Reverse the proportion

$$a : b = c : d$$

to

$$c : d = a : b$$

and apply the less-than equivalence already established.

Thus the third comparison form is a direct consequence of symmetry.

11 Formal Statement

The final production theorem is:

```

theorem euclid_proposition_5_14
  [HilbertIncidence Geo]
  [HilbertArchimedeanPlane Geo]
  (a b c d : HilbertPositiveSegmentClass Geo)
  (hProp : HilbertEudoxusProportion Geo a b c d) :
  (HilbertPositiveSegmentLess Geo a c <->
    HilbertPositiveSegmentLess Geo b d) /\
  (a = c <-> b = d) /\
  (HilbertPositiveSegmentLess Geo c a <->
    HilbertPositiveSegmentLess Geo d b) := by

exact
  \langle hilbertEudoxus_v14_lt_iff
    Geo a b c d hProp,
  hilbertEudoxus_v14_eq_iff
    Geo a b c d hProp,
  hilbertEudoxus_v14_gt_iff
    Geo a b c d hProp\rangle

```

The final wrapper simply packages the three comparison theorems.

12 Lean Proof Architecture

The production file contains five proposition-local components:

```

hilbertEudoxus_v14_lt_second_of_lt_first
hilbertEudoxus_v14_eq_second_of_eq_first

```

```

hilbertEudoxus_v14_lt_iff
hilbertEudoxus_v14_eq_iff
hilbertEudoxus_v14_gt_iff

```

followed by

```

euclid_proposition_5_14

```

which packages them.

The architecture is intentionally layered. The difficult mathematical input is isolated in the first strict direction; the other comparison forms are then derived by trichotomy and symmetry.

13 Axiomatic Strength

The proposition is stated under

```

[HilbertIncidence Geo]
[HilbertArchimedeanPlane Geo]

```

The Archimedean assumption enters indirectly through V.8.

V.13 and V.10 themselves require only the weaker congruence-level magnitude infrastructure, but V.14 depends on V.8 to generate the strict comparison that starts the proof.

Thus the proposition-level assumption records the strongest dependency in the actual DAG rather than suggesting that every local step is Archimedean.

14 Relationship with V.8, V.10, and V.13

V.14 is the first proposition in the current Book V reconstruction whose formal module imports other Euclid-numbered Book V proposition files:

```

import CGJteamLab.Proposition5_8
import CGJteamLab.Proposition5_10
import CGJteamLab.Proposition5_13

```

This is deliberate.

The dependency graph is not hidden inside a monolithic interface file. Instead, the module structure records the mathematical proposition graph:

$$V.8 \longrightarrow V.13 \longrightarrow V.10 \longrightarrow V.14.$$

The actual data flow is:

$$a < c \xRightarrow{V.8} c : d > a : d \xRightarrow{V.13} a : b > a : d \xRightarrow{V.10} b < d.$$

15 Relationship with V.16

V.14 is the decisive order-transfer theorem used in V.16 alternando.

To prove

$$a : c = b : d$$

from

$$a : b = c : d,$$

V.16 fixes arbitrary positive-multiple indices m, n and first constructs an equality of ratios

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

V.14 then gives all three comparison equivalences:

$$M_m(a) < M_n(c) \iff M_m(b) < M_n(d),$$

$$M_m(a) = M_n(c) \iff M_m(b) = M_n(d),$$

and

$$M_n(c) < M_m(a) \iff M_n(d) < M_m(b).$$

Those three clauses are exactly Definition 5 for the alternated ratio.

Thus V.14 is the order-theoretic heart of the formal proof of V.16.

16 Connection with the XI.23 Program

The original motivation for formalizing this fragment of Book V came from Euclid XI.23.

The classical XI.23 argument reaches a proportion of the form

$$OL : OP = LM : PQ.$$

If

$$OP < OL,$$

then the comparison-preservation content supplied by the Book V layer gives the corresponding comparison

$$PQ < LM.$$

In the present architecture this kind of transfer is encoded by the same Eudoxean comparison machinery that culminates in V.14 and V.16.

Thus the Book V reconstruction is not detached infrastructure: it supplies the exact ratio-order mechanism needed by the later solid-geometry proof.

17 Hidden Formal Data

Several features invisible in the printed ratio notation become explicit in Lean.

17.1 Three comparison forms are separate theorems

Although mathematically they are parts of one trichotomy statement, the formal development proves the $<$, $=$, and reverse- $<$ forms in a controlled order so that later proofs can reuse the exact direction needed.

17.2 Symmetry of proportion is explicit

Switching

$$a : b = c : d$$

to

$$c : d = a : b$$

is performed by

```
hilbertEudoxusProportion_symm
```

rather than being treated as syntactic interchange.

17.3 Strict and equal ratio relations are incompatible

The equality case relies on the reusable theorem

```
hilbertEudoxusProportion_not_greater
```

which formalizes the fact that ratios equal in the sense of Definition 5 cannot simultaneously satisfy Definition 7 in the same direction.

18 Classical DAG versus Formal DAG

The central strict direction has the proposition graph

$$\begin{array}{c}
a < c \\
\Downarrow \text{ V.8} \\
c : d > a : d \\
\Downarrow \text{ V.13} \quad \text{using } a : b = c : d \\
a : b > a : d \\
\Downarrow \text{ V.10} \\
b < d.
\end{array}$$

The complete formal proposition then adds:

$$\begin{array}{c}
a = c \\
\Downarrow \\
\text{trichotomy on } b, d \\
\Downarrow \\
b = d,
\end{array}$$

and uses symmetry plus trichotomy to derive the converse and reverse strict forms.

At module level the same graph appears through imports:

$$\begin{array}{c}
\text{Proposition5_8} \\
\text{Proposition5_10} \\
\text{Proposition5_13} \\
\Downarrow \\
\text{Proposition5_14.}
\end{array}$$

19 Structural Lesson

V.14 shows why preserving the proposition-level DAG matters.

One could hide V.8, V.10, and V.13 as anonymous helpers inside a large Book V interface and prove V.14 from them. The final theorem would compile, but the Euclidean structure would disappear.

The current organization keeps two layers distinct:

- `HilbertInterfaceV.lean` contains reusable magnitude and Eudoxean infrastructure;
- `Proposition5_n.lean` files expose Euclid’s numbered theorem graph.

V.14 is the first place where this separation becomes visibly valuable: the Lean import graph mirrors the mathematical proof graph.

20 Conclusion

Proposition V.14 is the comparison-preservation theorem for equal Eudoxean ratios.

Its strict core is the three-step chain

$$\text{V.8} \longrightarrow \text{V.13} \longrightarrow \text{V.10},$$

while equality and reverse strict comparison are derived by trichotomy and symmetry.

The proposition packages the complete statement

$$a < c \iff b < d, \quad a = c \iff b = d, \quad c < a \iff d < b,$$

and is the decisive order-transfer tool later used in V.16 alternando and, through that theorem, in the classical Book XI program.